#school/s2026/4-ApCalcBC

Types of Series

Series	Looks Like	Converges or Diverges
Geometric Series	$\sum_{n=1}^{\infty} a_0 r^n$ OR $\sum_{n=1}^{\infty} a_1 r^{n-1}$	$\|\|r\|\| \geq 1 \rightarrow$ Diverges $\|\|r\|\| < 1 \rightarrow$ Converges
P-Series	$\sum_{n=1}^{\infty} \frac{1}{n^p}$	$p > 1 \rightarrow$ Converges $p \leq 1 \rightarrow$ Diverges
Harmonic Series	$\sum_{n=1}^{\infty} \frac{1}{n}$	Diverges

Convergence Tests

Test	Conditions	Converges or Diverges
Nth Term Test	N/A	$\lim_{n\rightarrow\infty} a_n \neq 0 \rightarrow$ Diverges $\lim_{n\rightarrow\infty} a_n = 0 \rightarrow$ Inconclusive
Comparison Test	Positive, $a_n < b_n$	if a_n Diverges, then b_n Diverges if b_n Converges, then a_n Converges
Limit Comparison Test	Positive	if $\lim_{n\rightarrow\infty} \frac{a_n}{b_n} = L$ where L is finite and positive, both a_n and b_n converge or diverge
Integral Test	Positive, Decreasing, Continous for $x = k$ $a_n = f(x)$	$\int_k^{\infty} f(x)$ and $\sum_{n=k}^{\infty} a_n$ either both converge or both diverge
Alternating Series Test	Alternating $(-1)^n$	1) $\lim_{n\rightarrow\infty} a_n = 0$ 2) $\|\|a_n\|\| \geq \|\|a_{n+1}\|\|$ (Decreasing) Else: Inconclusive(Diverges for the sake of this class)

Test	Conditions	Converges or Diverges
Ratio Test	Positive, Not a P-series	$\lim_{n \to \infty} \frac{a_{n+1}}{a_n} > 1 \to$ Diverges $\lim_{n \to \infty} \frac{a_{n+1}}{a_n} < 1 \to$ Converges $\lim_{n \to \infty} \frac{a_{n+1}}{a_n} = 1 \to$ Inconclusive